

MTH 204: Lecture 01

Modeling:

Physical System



Mathematical Model

→ ODEs/PDEs



Mathematical solution



Physical Interpretation

A mathematical model is one or more equations that helps us to understand a physical process.

- A first order ODE is an equation of the form:

$$F(y', y, x) = 0$$

Any function involving y' , y and x

$x \rightarrow$ independent variable (time, distance)

$y \rightarrow$ dependent variable $y(x)$

$y' \rightarrow$ First derivative of y i.e. $y' = \frac{dy}{dx}$

(ordinary derivative)

Example: $y' = \cos x$ (1st order)

2nd order ODE: $F(y'', y', y, x) = 0$

3rd order ODE: $G(y''', y'', y', y, x) = 0$

Examples: $y'' + 9y = e^{-2x}$ (2nd order)

$y' y''' - \frac{3}{2} (y')^2 = 0$ (3rd order)

Order: Highest derivative of dependent variable present in the equation determines the order.

ODE: It is an equation involving

- One independent variable (x)
- One dependent variable (y)
- Derivatives of dependent variable

(of different orders) w.r.t. the independent variable
(ie. $y', y'', y''' \dots$)

- Must contain a derivative term.

Ex:

$$y' = \cos x$$

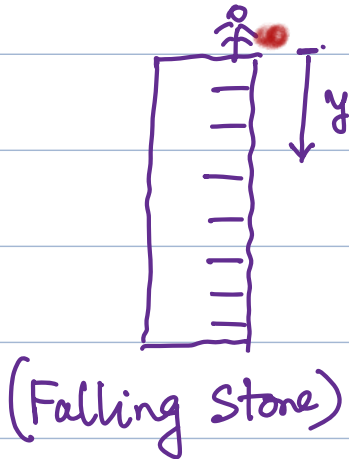
$$\Leftrightarrow y = \sin x + C \rightarrow \text{Not a Differential Equation.}$$

$$y'' = -\sin x.$$

Examples:

$$m y'' = mg$$

①



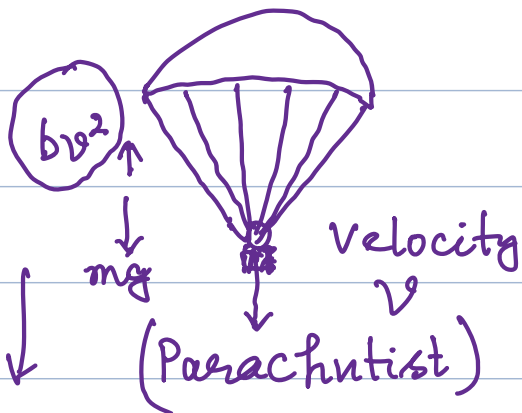
Independent variable: t

Dependent variable: y

$$\frac{d^2 y}{dt^2} = y'' = g = \text{constant}$$

$$\left(y = \frac{1}{2} g t^2 \right)$$

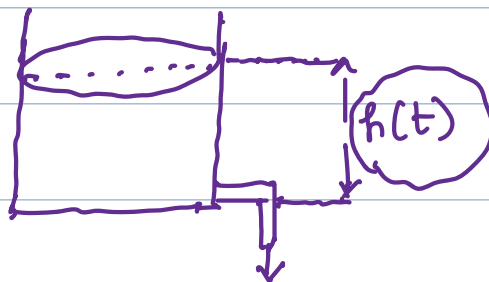
②



$$m \frac{dv}{dt} = m v' = mg - bv^2$$

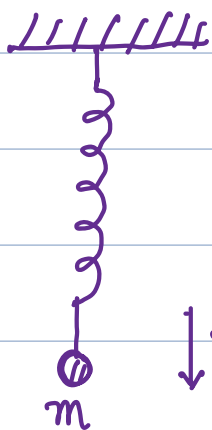
$b \rightarrow$ "drag" constant

③



$$\frac{dh}{dt} = h' = -k\sqrt{h}$$

4



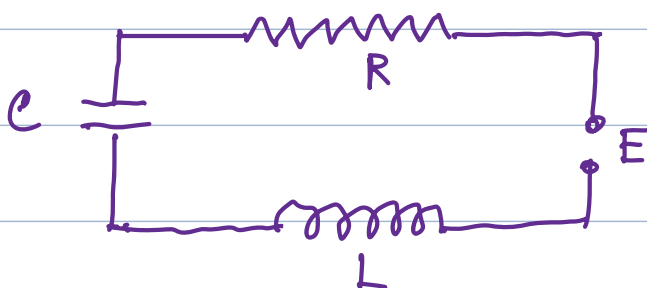
Displacement = y

$$my'' + ky = 0$$

(Vibrating mass on a spring)

k = Spring Constant.

5

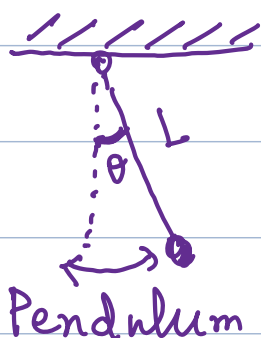


Current I in a RLC Circuit

$$LI'' + RI' + \frac{1}{C}I = E'$$

independent variable = t (time)

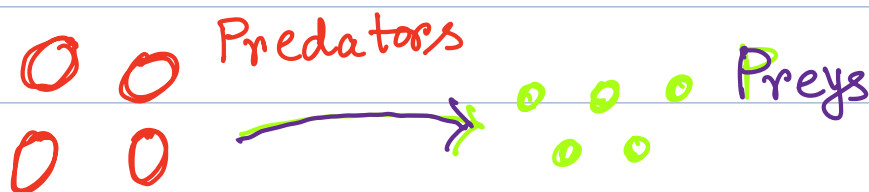
6



$$L\theta'' + g \sin \theta = 0$$

independent variable = t time)

7



(Lotka - Volterra predator-prey model)

$$\begin{aligned} y_1' &= ay_1 - by_1y_2 \\ y_2' &= ky_1y_2 - ly_2 \end{aligned} \quad \left(\begin{array}{l} \text{independent variable} \\ = t \end{array} \right)$$

(System of two first order ODEs)

ODEs

$$f(x), \frac{df}{dx}, \frac{d^2f}{dx^2}, \dots$$

- Highest derivative present is the order

PDEs

$$f(x, y), \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}, \frac{\partial^2 f}{\partial x^2}, \frac{\partial^2 f}{\partial x \partial y}, \dots$$

(Both x & y are independent variable)

- Highest partial derivative present in the equation determines the order

Definition:

Solution: A function $y = h(x)$ is a solution of a given ODE $F(x, y, y', \dots, y^{(n)}) = 0$ on some open interval $a < x < b$ (a, b) if $h(x)$ is defined and it has derivatives upto n th order ($h, h', \dots, h^{(n)}$) in the interval (a, b) and the equation is satisfied if $y, y', y'', \dots, y^{(n)}$ are replaced

by $h, h', h'', \dots, h^{(n)}$

i.e. $\boxed{F(x, h, h', \dots, h^{(n)}) = 0}$ on $a < x < b$

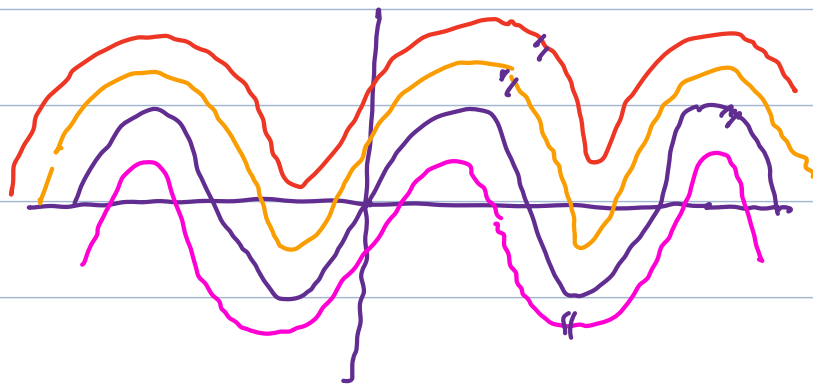
Ex: Consider the ODE: $xy' = -y$

Then $y = \frac{c}{x}$ is a solution on $(0, \infty)$ or on $(-\infty, 0)$
because $y' = -\frac{c}{x^2} \Rightarrow xy' = x \cdot \left(-\frac{c}{x^2}\right) = -\frac{c}{x} = -y$

Thus $h(x) = \frac{c}{x}$

Ex: $y' = \cos x$

Then $y = \int \cos x \, dx$
 $\Rightarrow y = \sin x + C$
on $(-\infty, \infty)$



Ex: $y' = ky$, k is a constant

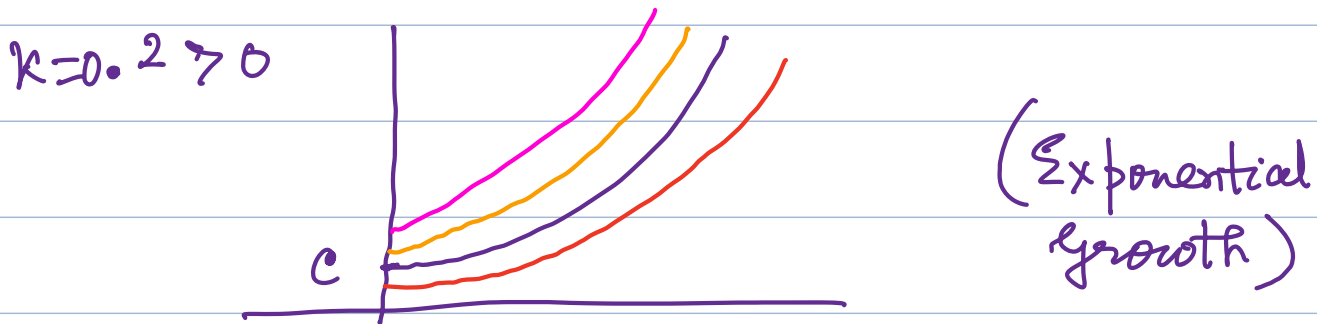
$$\Rightarrow \frac{dy}{dx} = ky \Rightarrow \int \frac{dy}{y} = \int k \, dx$$

$$\Rightarrow \ln|y| = kx + c_1 \text{ where } c_1 \text{ is a constant}$$

$$\Rightarrow y = e^{kx+c_1} = e^{c_1} e^{kx}$$

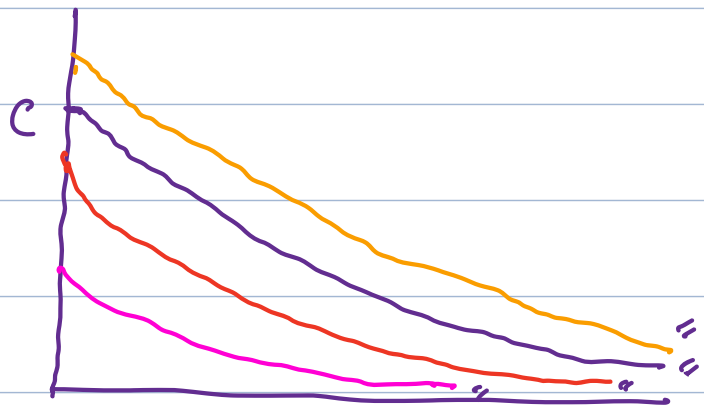
$$\Rightarrow \boxed{y = c e^{kx}} \text{ where } c = e^{c_1}$$

If $k > 0$, it will give exponential growth }
 If $k < 0$, it will give exponential decay }



$k = -0.2 < 0$

(Exponential Decay)



General Solution:

$$y = c e^{kx}$$

$c = \text{arbitrary constant.}$

Particular Solution:

$$y = c e^{kx} \quad y(0) = 5 \Rightarrow c = 5 \quad y = 5 e^{kx}$$

Initial Value Problem (IVP):

$$\left. \begin{aligned} y' &= f(x, y) \\ y(x_0) &= y_0 \end{aligned} \right\}$$

In the above case $y' = ky$, $y(0) = 5$

Ex: IVP : $y' = 3y$, $y(0) = 5.7$

General solution : $y = Ce^{3x}$

Now $y(0) = 5.7 \Rightarrow 5.7 = Ce^{3 \times 0} \Rightarrow C = 5.7$

Hence $\boxed{y = 5.7 e^{3x}}$ $\rightarrow$ Particular solution of the IVP

Modeling: Radioactivity :

Step 1: Setting up a mathematical model

$y(t) \rightarrow$ quantity of a radioactive substance after time t

Then $\boxed{y' = -ky}$

$k \rightarrow$ depends on the substance

Unit of $k \rightarrow s^{-1}$

$k = 1.4 \times 10^{-11} s^{-1}$ for

Ra_{88}^{226} (Radium)

Step 2: Find solution of mathematical model

$$y(t) = e e^{-kt}$$

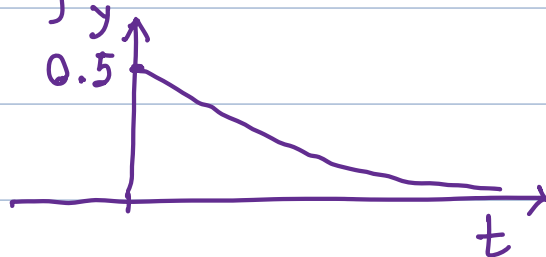
(given $y(0) = 0.5 \text{ kg}$)

Then $y(t) = 0.5 e^{-kt}$

Step 3: Interpretation of solution

$$k = 1.5$$

$$y(t) = 0.5 e^{-1.5t}$$



Half Life: $y(T) = \frac{y(0)}{2}$ Find T ?

$$0.5 e^{-1.5T} = \frac{0.5}{2} \Rightarrow e^{-1.5T} = \frac{1}{2} \Rightarrow -1.5T = -\ln 2 \Rightarrow \frac{3}{2}T = \ln 2 \Rightarrow \boxed{T = \frac{2 \ln 2}{3}}$$

Geometric Meaning:

$$F(y', y, x) = 0$$

→ Most general form of
a first order ODE

$$y' = f(x, y) \rightarrow \text{Explicit form of a first order ODE}$$

↓
 $\frac{dy}{dx}$ is given by $f(x, y)$

→ slope of the tangent at the point
 $(x, y(x))$ of the curve $y = y(x)$

Ex: $y' = x + y$ So, $f(x, y) = x + y$

At each point (x, y) of the plane ($\mathbb{R}^2$)

we can draw a small vector representing
the slope $x + y$.

(x, y)	$x+y$
$(0, 1)$	1
$(1, 0)$	1
$(1, 1)$	2
$(-1, 0)$	-1
$(0, -1)$	-1
$(1, 2)$	3
$(2, 1)$	3



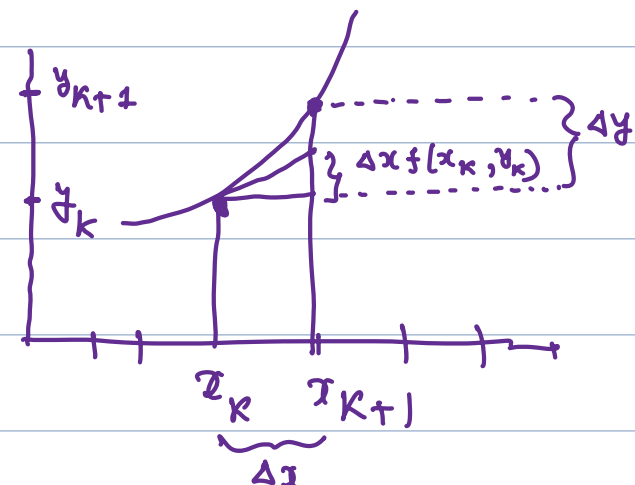
Direction field of $y' = x + y$

- Vectors in direction field represent tangent vectors of solution curves

Euler Method (Numerical).

$$y' = f(x, y)$$

$$\frac{\Delta y}{\Delta x} \approx \frac{dy}{dx} = f(x, y)$$



Let $\Delta x = x_{k+1} - x_k$

and $\Delta y = y_{k+1} - y_k$

Then $\Delta y \approx \Delta x f(x, y)$

$\Rightarrow y_{k+1} - y_k \approx \Delta x f(x, y)$

$\Rightarrow y_{k+1} \approx y_k + \Delta x f(x_k, y_k)$

Let $\Delta x = h$ (fixed)

(x_0, y_0)

(x_1, y_1)

(x_2, y_2)

(x_3, y_3)

$\vdots$

$$\begin{cases} x_1 = x_0 + h, & y_1 = y_0 + h f(x_0, y_0) \\ x_2 = x_1 + h, & y_2 = y_1 + h f(x_1, y_1) \\ x_3 = x_2 + h, & y_3 = y_2 + h f(x_2, y_2) \\ \vdots & \vdots \end{cases}$$

Separable ODEs :

A First order ODE is separable if it can be written as

$$g(y) y' = f(x)$$

(y on the left)

(x on the right)

Then $g(y) y' = f(x)$

$$g(y) dy = f(x) dx$$

$$\Rightarrow \int g(y) dy = \int f(x) dx + C$$

Ex: $y' = 1 + y^2$

$$\frac{dy}{1+y^2} = dx$$

$$\int \frac{dy}{1+y^2} = \int dx \Rightarrow \tan^{-1} y = x + C$$

$$\Rightarrow \boxed{y = \tan(x + C)}$$

Ex: $y' = (x+1) e^{-x} y^2$

$$\int \frac{dy}{y^2} = \int (x+1) e^{-x} dx$$

$$\Rightarrow -\frac{1}{y} = -(x+1)e^{-x} - e^{-x} + C_1$$

$$\Rightarrow -\frac{1}{y} = -x e^{-x} - 2e^{-x} + C_1$$

$$\Rightarrow -\frac{1}{y} = -(x+2)e^{-x} - C$$

$$\Rightarrow \frac{1}{y} = (x+2)e^{-x} + C$$

$$\Rightarrow \boxed{y = \frac{1}{(x+2)e^{-x} + C}}$$

Ex: $y' \sin(2\pi x) = \pi y \cos(2\pi x)$

$$\Rightarrow \frac{dy}{y} = \pi \frac{\cos(2\pi x)}{\sin(2\pi x)} dx$$

$$\Rightarrow \int \frac{dy}{y} = \int \pi \frac{\cos(2\pi x)}{\sin(2\pi x)} dx$$

For the integral on the R.H.S., let $u = \sin(2\pi x)$

$$\Rightarrow du = 2\pi \cos(2\pi x) dx$$

$$\Rightarrow \frac{1}{2} du = \pi \cos(2\pi x) dx$$

Then $\int \frac{dy}{y} = \frac{1}{2} \int \frac{du}{u}$

$$\Rightarrow \ln|y| = \frac{1}{2} \ln|u| + C_1$$

$$\Rightarrow \ln|y| = \frac{1}{2} \ln|\sin(2\pi x)| + C_1$$

$$\Rightarrow \ln|y| = \ln \sqrt{|\sin 2\pi x|} + C_1$$

$$\Rightarrow |y| = e^{C_1} \sqrt{|\sin 2\pi x|}$$

$$\Rightarrow \boxed{|y| = C \sqrt{|\sin 2\pi x|}}$$